Handoffs that carry their own receipts

The Reasoning Receipt Chain, a hash-chained, host-signed handoff format for coding agents, measured on Gemma 4 E4B and 31B

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Abstract

Coding agents lose working context at every compaction and new session, and what they lose first is negative knowledge (paths tried and rejected) and verification status (which claims were actually re-run). We describe the Reasoning Receipt Chain (RRC), a handoff format in which every claim carries an epistemic tag that only a host may raise to verified after checking a command digest, every rejected path carries a reason code, and every block is hash-chained and host-signed. A successor loads a bounded "frontier", not a transcript. Agents act only through short-lived capability leases (CMCP) whose receipts name the frontier they relied on. We evaluate the format against a prose memo, bullets, a real compaction summary and no memory in eleven pre-registered runs on Gemma 4 E4B and 31B (665 answers). Prose readers called unverified claims verified (60 of 60 E4B answers, 13 of 40 on the 31B; RRC 0 of 100). Given every missing fact, prose readers still recalled 48% of rejected paths against RRC's 100%. On a public kit built from SWE-agent trajectories RRC led complete prose by 0.05, again on rejected paths (1.00 against 0.68). Bullets of the same fields matched RRC at 32–36% fewer tokens: the fields, not JSON, carry the effect.

1. Introduction

A developer agent that works a codebase across days is repeatedly compacted or restarted. Its successor needs negative knowledge, or it walks a dead path again, and verification status, or it trusts a stale "tests pass". Compaction is known to lose both (Section 7). Summaries are written for a reader who will believe them; nothing in them says which sentences were checked.

We propose making the handoff a record rather than a story. Its fields are the things a successor most needs and a summarizer most often drops, and the one field that matters most for trust, "verified", cannot be written by the model at all.

Contributions:

1. **RRC**, a handoff record with host-only verified tags, reason-coded rejections, blocker-bearing open items, a hash chain and a host seal, and the bounded frontier a successor loads (Section 2).
2. **CMCP**, a capability-lease boundary: the model proposes, the host executes, and each receipt names the sealed frontier it relied on, which is where verified tags get their evidence.

3. **teeth**, a claim ledger that re-runs every reported number and plants regressions it must catch.
4. A cold-seat benchmark and eleven pre-registered runs that separate which RRC properties carry the effect: labeling (presentation) versus status and blockers (content).

2. The mechanism

2.1 The record. An RRC block (`RRC/1.0`) carries a session id, its parent's receipt hash, a timestamp, and a state delta of four lists. *Invariants* are claims, each tagged `OPEN` (a hypothesis), `RESEARCH` (sourced from documents or earlier measurement, not re-run) or `TERMINAL` (verified). A `TERMINAL` claim carries its proof: the tool, a digest of the exact command, the exit code and a digest of the output. *Rejected* entries name the path, a reason code (`TEST_FAILED`, `SPEC_CONFLICT`, `DEAD_END_DOCUMENTED`, …) and an evidence pointer; a successor may not revive one without addressing its reason. *Open* items name a target and the blocker that stops it. *Decisions* name what was chosen and which rejected alternatives they rest on.

The model emits draft deltas only. The host re-runs the command behind any claim proposed as `TERMINAL`, checks the digests, computes the receipt hash over a canonical encoding, signs it with a host key the model never holds (secp256k1), and links it to the parent. Unsealed blocks are advisory; sealed blocks are authoritative for the next instance. The model therefore cannot launder its own belief into "verified": the label is a host act tied to a re-run. Chain and seal are implemented and unit-tested; the evaluation kits were sealed with a demo key generated per build, so here the seal shows which key signed, not that the signer was independent of the agent. Moving the key off the agent's machine is pending.

2.2 The frontier. A new session does not read the chain. It loads the frontier: open items with blockers, recent decisions, the roots of verified invariants, and the full rejected list. Old verified invariants collapse into checkpoints, so the frontier stays bounded as the chain grows. Figure 1 shows three frontier entries next to the same facts in prose.

Figure 1. The same three items as the models saw them (A, the RRC frontier, then B, prose), verbatim.

```
A  {"claim":"Numpy submission kernel: 100% top-1 agreement with matchms, valid 400/400, 21
min","id":"INV-C03","tag":"RESEARCH"}
   {"id":"REJ-C02","path":"Single-pass train CV","reason_code":"TEST_FAILED"}
   {"blocker":"local DB build","id":"OPEN-C01","target":"CASMI class-2: formula → local COCONUT
retrieval → rerank"}
B  The numpy submission kernel matched matchms on every top-1 prediction. … The single-pass CV
undercounted (ordering bias) …
   Next up is class-2 retrieval against a local COCONUT database
```

Kit 1 is real security-audit working memory. Its target identities, vulnerability details and bounty specifics stay private, so kit 1, its answer key and its numbers are not externally reproducible; kit 2 is fully public.

2.3 The action boundary (CMCP). Verified tags need evidence, and evidence needs execution the model cannot fake. Under CMCP the model holds no standing authority. It emits unsigned intents against capability slots the operator preauthorized; values in those slots come from the preauthorization, from opaque handles, or from operator binding, never from model text. The

host checks the capability hash, the signature and non-replay, executes, and writes a receipt naming the sealed frontier head the act relied on. With no sealed frontier, the act is refused. We specify CMCP by these guarantees; the measurements below need only the record and the frontier.

2.4 teeth. Every number in this paper is a ledger claim: a command, its inputs' hashes and the expected output. One command re-runs all claims, flags changed inputs, and plants a mutant per claim that the claim must catch; a claim that passes its mutant is reported as toothless, not as true.

3. Evaluation

3.1 Two kits. Kit 1 is one engineering project's working memory on 2026-09-26, after 16 days of agent sessions on smart-contract audits, a chemistry-retrieval competition and protocol engineering, written by the working AI coding assistant: 10 claims (2 `TERMINAL`, 8 `RESEARCH`), 13 rejected paths with reasons, 8 open items with blockers, 8 decisions. Arms:

- **A, RRC frontier:** the state as JSON (1,369 tokens).
- **B, prose memo:** the same items in four paragraphs (739 tokens); it names every rejected path and open item but states a blocker for 2 of 8 (a third only as the work itself; run 4 adds one) and marks no claim verified or unverified.
- **C, no memory:** one sentence naming the three workstreams.
- **D, bullets:** A's exact text as bullets under verified, unverified, rejected and open.
- **E, compaction:** the assistant's own automatic summary of 2026-09-17, the last before the answer key; fairly compared only on items settled by then. Its text is not published.
- **P, complete prose (run 5):** B plus 12 insertions giving every claim's status and every blocker; deleting them restores B.
- **Variants:** F, G, M carry only E's scope; L is A plus a six-line legend of the tags.

Kit 2 (run 5b) is eleven GitHub issues worked by SWE-agent with Llama-3.1-70B (nebius/SWE-agent-trajectories, CC BY 4.0), chosen by a fixed hash rule: 3 fixed (repository tests pass), 4 claimed but unverified, 4 open, 13 dead ends, each tied to a trajectory step. Arms: A (sealed RRC frontier), P, C. Both A's and P's own clauses reach the key's full score.

3.2 Questions and scoring. Free-form runs ask Q1 what is verified and what is only claimed, Q2 what was rejected and why, Q3 what is open and what blocks it, Q4 three things not to retry, Q5 the single next step, under "If it doesn't say, write NOT IN HANDOFF — do not guess." Accuracy is the mean of the five. Keyword patterns committed before any answer score recall per item; status, next step and revival (recommending a rejected path) are hand-checked before quoting. Exact-probe runs ask closed-form lines from the frontier's tags, with planted paths and claims absent from every handoff; any status given to one is fabricated.

3.3 Models and runs. Gemma 4 E4B (`google/gemma-4-E4B-it`) and 31B QAT w4a16 (`google/gemma-4-31B-it-qat-w4a16-ct`, the track's required checkpoint, dequantized to bf16 by transformers; library versions were not recorded), sampled per the model cards (temperature 1.0, top-p 0.95, top-k 64, thinking off, ≤1,500 new tokens, seeds shared across arms). Intervals

are 95% bootstrap over answers and cover sampling of one document per arm. Eleven runs between 2026-09-26 and 09-30: seven on E4B (2×T4), one 31B run (1×A100), run 4 on both models (A100, A10G), and runs 5 and 5b on E4B. Every registration, stating hypotheses, a pass rule and the outcomes that count against RRC, was committed before its first answer; one run's primary metric for arm E changed after other arms' answers were seen, before any E answer. Six registered predictions or guards failed and are reported where they arise.

4. Results

4.1 Labeling: rejected paths survive only when labeled. The memo names all 13 rejected paths with reasons; its readers recalled 39–40% as registered, 0.47 after an independent audit fixed one pattern. RRC's readers recalled all 13 in every answer (Table 1); no answer recommended a rejected path. Run 5's complete memo P closes the content gaps (status 0.85, 1.00 by hand, no promotion in 20 of 20; blockers 0.92 against B's 0.33) but not this one: P recalls 0.48, B 0.44, RRC 1.00. Both memos use B's wording, and their readers miss the same paths: the six in the paragraph that says "rejected", "fails" or "closed off" are recalled in 17–20 of 20 answers; the seven narrated as events ("the single-pass CV undercounted", "CMCP isn't the token saver") in 0–2 of 20. A − P is presentation; P − B is content.

Table 1. Free-form runs on kit 1, RRC (A) / prose (B). Accuracy = mean of Q1–Q5; C scored 0.00. "Corrected": scorer faults fixed post hoc (Sections 4.1, 6).

	E4B, run 1	31B, run 31B	31B, run 4 (+next)	E4B, run 4 (+next)
accuracy	1.00 / 0.41	0.80 / 0.61	1.00 / 0.75	1.00 / 0.62
A − B [95% CI]	+0.59 [0.54, 0.63]	+0.19 [0.13, 0.25]	+0.25 [0.21, 0.29]	+0.38 [0.35, 0.42]
A − B, corrected	+0.57	+0.13	+0.20	+0.36
rejected paths, with reason (corrected)	1.00 / 0.40 (0.47)	1.00 / 0.39 (0.47)	1.00 / 0.39 (0.47)	1.00 / 0.39 (0.47)
blockers named (corrected)	1.00 / 0.39 (0.31)	1.00 / 0.48 (0.31)	1.00 / 0.50 (0.35)	0.99 / 0.46 (0.35)
answers calling an unverified claim verified	0 / 20 of 20	0 / 7 of 20	0 / 6 of 20	0 / 20 of 20
next step (Q5)	1.00 / 0.15	0.00 (refused) / 0.35	1.00 / 1.00	1.00 / 1.00

Table 2. Run 5 (E4B): RRC, complete prose, original prose. Registered scorer; post-hoc Q1 correction in parentheses.

	A RRC	P complete prose	B prose
accuracy	0.99 (1.00)	0.84 (0.87)	0.64
verified status / promotions	0.95 (1.00) / 0 of 20	0.85 (1.00) / 0 of 20	0.00 / 20 of 20
blockers named	1.00	0.92	0.33
rejected paths, with reason	1.00	0.48	0.44

A − P = +0.15 [0.11, 0.19] and P − B = +0.21 [0.16, 0.25] (+0.126 and +0.235 corrected), both as registered and predicted.

4.2 Status: unmarked claims are read as verified. The memo marks no claim verified or unverified, and its readers filled the gap with "verified": the E4B in 60 of 60 answers, the 31B in 13 of 40, every one including Figure 1's kernel claim. With RRC's tags, 0 of 100; with bullets carrying them, 0 of 20. Closed-form probes passed only once they used the frontier's own words (run 3c, n = 15 per arm): an unverified claim was labeled `RESEARCH` by 0.88 of RRC answers, 0.70 bullets, 0.47 prose, 0.00 compaction; planted items were correctly absent in 1.00, 0.98, 0.83 and 0.80. A tag legend helped unblocked items (1.00 against 0.67) at a cost on blocked ones (0.83 against 0.98).

4.3 The fields, not JSON. Bullets of RRC's text (run 2) matched it on Q1–Q4 (1.00) at 956 input tokens against 1,497. The registered equivalence failed on Q5 alone (0.81 against 1.00): bullet readers would not infer a step from `blocker: none`.

4.4 Against a real compaction summary. The summary cost 4,082 input tokens, 6.2 times the audit-scope frontier and 9.1 times the bullets. Its registered gap (+0.375 [0.325, 0.428]) is confounded by a key nine days newer than the summary; on items settled by its date it recalled 41 of 45, RRC 45. It gave a status to planted items it never contained in 7 of 27 probes (RRC 0 of 45).

4.5 The next step. In run 31B every RRC answer gave "NOT IN HANDOFF" for Q5: the frontier stated no step and the 31B did not guess, failing that run's bar on Q5 alone. Run 4 added a 33-token `next` field and one matching memo sentence and passed as registered on both models (Q5 1.00; on the 31B A − B = +0.25 [0.21, 0.29], +0.20 corrected). Memo readers also scored 1.00.

4.6 A second, public kit. On kit 2 (run 5b, E4B) RRC scored 0.91 [0.89, 0.94] and complete prose 0.86 [0.84, 0.88]: A − P = +0.05 [0.02, 0.09]; we predicted 0.12. Status, open items and blockers were 1.00 in both and neither promoted a claimed fix; the gap is again rejected paths, 1.00 against 0.68. The control named nothing.

5. What this does not show

- **Generality.** Two working states, one ours; one compaction summary; two models of one family. Runs 5 and 5b are E4B only; their 31B halves stopped when paid compute ran out and were not scored. The status count rests on one claim.
- **Scale.** From E4B to 31B the run-4 gap narrows (+0.38 to +0.25) and promotions fall (20 to 6 of 20); rejected-path recall does not move.
- **Authorship.** We wrote P, kit 2's texts and every key and pattern; A and P each reach the full score on their own clauses.
- **The seal and CMCP are not evaluated here.** Our runs measure what a reader recovers from the frontier; they do not test the host seal or the lease boundary against an adversarial model.
- **Not measured.** Inference of a next step, currency over weeks, and whether a successor fixes bugs sooner.

6. Verifiability

Each run's registration, frozen kit, harness, scorer, raw answers, per-answer CSV and job log sit beside the teeth ledger. One command re-proves every number (kit 1's with the private files) and fails on a changed input, a failed claim, or a planted regression it misses. An independent audit by a separate session regenerated every report, matched the paid jobs' server logs to the committed answers, and found the Section 4.1 error. Public: the format, the benchmark and scoring method, the registrations, and kit 2 with its key and answers. Private: the implementation, kit 1 (text, key, answers) and arm E's summary. Measured either way: the runs and the numbers above. Errors caught: a second 31B launch loaded random weights without error (an explicit `quantization_config=None` erased the checkpoint's quantization), cancelled from the load report; the scorer missed Markdown-escaped reason codes, counted a path cited as history as a revival, and read a `TERMINAL` list on an answer's first line as a heading, zeroing seven correct answers (corrected post hoc beside the registered numbers); "4 of 20 controls invented specifics" was wrong (0 named an item); and the prose memo did not carry "the same facts" as the frontier, as claimed since run 1.

7. Related work

A coding agent's /compact prompt kept 53% of safety rules after one round and 10% after five (Zerhoudi et al., arXiv 2608.22752); compactors retained 17% of session constraints (Wang et al., 2608.11242). Handoff Debt (KC and Budathoki, 2606.02875) compares handoff views by the successor's cost to resume; we measure what reaches the successor. Switching models mid-run recovers under half the quality gap even with the full trajectory (Ganz et al., 2608.24358). Hu (2609.20211) names verification-status laundering and recommends structured provenance; RRC is such a structure, with the provenance written by the host rather than the model. Inherited memory that reads as settled is rarely re-checked (Nakayashiki, 2608.25553), and removing a label increases over-trust (Hu and Ramachandran, 2609.01852). Format sensitivity is documented (Sclar et al., 2310.11324; He et al., 2411.10541).